

PYTHON DICTIONARY METHODS

> ADD / UPDATE

- Update() Method

Updates the existing key with new value. If the key does not exist, it will create a new key:value pair.

Example:

```
numbers = {1:'One', 2:'Two', 3:'Three'}  
numbers.update({3:'THREE'})  
numbers.update({4:'Four'})  
print(numbers)
```

Output:

```
{1:'One', 2:'Two', 3:'THREE', 4:'Four'}
```


> REMOVE

• pop() Method

Deletes item with the specified key from the dictionary.

Example:

```
numbers = {1: 'One', 2: 'Two', 3: 'Three'}
removed = numbers.pop(3)
print(numbers)
print(removed)
```

Output:

```
{1: 'One', 2: 'Two'}
Three
```

• popitem() Method

Removes and returns the last key-value pair from the dictionary.

Example:

```
numbers = {1: 'One', 2: 'Two', 3: 'Three'}
removed = numbers.popitem()
print(numbers)
print(removed)
```

Output:

```
{1: 'One', 2: 'Two'}
(3, 'Three')
```

• clear() Method

Removes all the elements from a dictionary.

Example:

```
numbers = {1:'One', 2:'Two', 3:'Three'}  
numbers.clear()  
print(numbers)
```

Output:

{}

> OTHERS

• copy() Method

Creates a duplicate copy of the dictionary.

Example:

```
numbers = {1:'One', 2:'Two', 3:'Three'}  
numbers2 = numbers.copy()  
print(numbers2)
```

Output:

{1:'One', 2:'Two', 3:'Three'}

• fromkeys() Method

Creates a dictionary from the listed keys and their value.

Example:

```
keys = (1, 2, 3)  
numbers = dict.fromkeys(keys)  
print(numbers)
```

Output:

{1: None, 2: None, 3: None}

- setdefaults() Method

Set and return the dictionary key with the default value.

Example:

```
numbers = {1:'One', 2:'Two', 3:'Three'}
value3 = numbers.setdefault(3, "THREE")
value4 = numbers.setdefault(4, "FOUR")
print(numbers)
print(value3)
print(value4)
```

Output:

```
{1:'One', 2:'Two', 3:'Three', 4:'FOUR'}
Three
FOUR
```

> ACCESS ITEMS

- get() Method

Using the get() Method, we do not get any key error if the key does not exist.

Example:

```
numbers = {1:'One', 2:'Two', 3:'Three'}
print(numbers.get(3))
print(numbers.get(4))
```

Output:

```
Three
None
```

- The items() Method

The items() Method returns the view of tuples, and each tuple contains the key and its value.

Example:

```
numbers = {1:'One', 2:'Two', 3:'Three'}  
print(numbers.items())
```

Output:

```
dict_items([(1, 'One'), (2, 'Two'), (3, 'Three')])
```

- keys() Method

The keys() Method returns the view of keys.

Example:

```
numbers = {1:'One', 2:'Two', 3:'Three'}  
print(numbers.keys())
```

Output:

```
dict_keys([1, 2, 3])
```

- values() Method

The values() method returns the view of values in dictionary.

Example:

```
numbers = {1:'One', 2:'Two', 3:'Three'}  
print(numbers.values())
```

Output:

```
dict_values(['One', 'Two', 'Three'])
```